Curriculum Vitae

Personal Data

Title	Prof. Dr. rer. nat.
First name	Matthias
Last name	König
Current position	Professor of Metabolic Inflammation and Carcinogenesis of the Liver (since 06/2026)
Current institution(s)/site(s), country	University of Lübeck and University Hospital Schleswig-Holstein (UKSH), Campus Lübeck, First Department of Medicine, Germany; livermetabolism.com
Identifiers/ORCID	ORCID: 0000-0003-1725-179X GitHub Google Scholar

Qualifications and Career

Stages	**Periods and Details**
Degree programme	Diploma in Biophysics (equivalent to MSc, grade 1.0 with distinction), 2002–2009, Humboldt-Universität zu Berlin, Germany. Diploma thesis on systems biology modelling of hepatic metabolism.
Doctorate	12/2015, Dr. rer. nat. in Biophysics (magna cum laude), Humboldt-Universität zu Berlin and Charité – Universitätsmedizin Berlin, Germany. Thesis: *Computational modeling of glucose metabolism*. Supervisor: Prof. Hermann-Georg Holzhütter.
Stages of academic/professional career	**since 06/2026** Professor of Metabolic Inflammation and Carcinogenesis of the Liver, University of Lübeck / UKSH, First Department of Medicine, Germany. Group *Systems Medicine and Digital Twins*. **05/2026** Associate Professor of Pharmacoepidemiology and Data Science, University of Southern Denmark (SDU), Odense, Denmark. **12/2025–04/2026** Head of the Computational Biomechanics Group, University of Stuttgart, Germany (coordination project of the DFG Priority Programme SPP 2311). **06/2021–11/2025** DFG Temporary Position for Principal Investigators (*Eigene Stelle*, FOR 5151 QuaLiPerF), head of the group *Systems Medicine of the Liver*, Institute for Theoretical Biology, Humboldt-Universität zu Berlin, Germany. **06/2016–05/2021** Junior group leader, BMBF LiSyM – Systems Medicine of the Liver, Institute for Theoretical Biology, Humboldt-Universität zu Berlin, Germany. **2008–2015** Research associate, Computational Systems Biochemistry, Institute of Biochemistry, Charité – Universitätsmedizin Berlin, Germany (BMBF consortia HepatoSys and Virtual Liver).

Activities in the Research System

2017–2026	**Elected COMBINE coordinator** – coordination of the community standards for computational models in biology (SBML, SED-ML, PEtab, SBGN, …) and of the annual COMBINE meetings.
2018–2023, 2025–2027	**Elected SBML editor** – Systems Biology Markup Language, exchange format for biological models.
2017–2022, 2024–2027	**Elected SED-ML editor** – Simulation Experiment Description Markup Language for reproducible simulation experiments.
2023–2026	**Elected PEtab editor** – data format for parameter estimation problems.
2024–2026	**Open Science Ambassador**, Berlin University Alliance / Humboldt-Universität zu Berlin.
2023–2024	**Expert panel member**, PharmVar CYP1A2 gene panel (Pharmacogene Variation Consortium).
2025	**Organiser** of the workshop *Open Science & Reproducibility for Computational Models in Systems Biology & Medicine*, Berlin.
2022	**Organiser** of the 13th COMBINE meeting, Berlin.
2022–2026	**Teaching and mentoring**: three BUA X-Student Research Groups on digital twins for drug therapy (hypertension, anticoagulation); certified *Digital Health Professions Educator* (Charité, 2025) and AI teaching certificate (HU Berlin, 2025).

Supervision of Researchers in Early Career Phases

Since 2021: 1 completed doctorate (J. Grzegorzewski, 2017–2023, *Physiologically based pharmacokinetic (PBPK) modeling for dynamical liver function tests and CYP phenotyping*, Humboldt-Universität zu Berlin), 4 Master's theses, 8 Bachelor's theses and 18 internship students. Currently supervised: 1 doctoral researcher (M. Myshkina, since 2024, AI and digital twins for decision support in hepatocellular carcinoma) and 1 medical doctoral thesis (A. Schwaiger, since 2023, PBPK/PD digital twin of hydrochlorothiazide). The Bachelor's theses of M. Elias, N. Nemitz and E. Tensil each resulted in a first-author, peer-reviewed digital twin publication (Category A, nos. 1–3).

Scientific Results

Total: 68 peer-reviewed publications. Own contribution is given in *italics*; all listed publications are related to the topic of TWIN-ORG.

Category A

1. Tensil E, Myshkina M, **König M** (2026). A digital twin of the angiotensin II receptor blocker losartan: physiologically based modeling of blood pressure regulation. *Pharmaceutics* 18(2):262. doi:10.3390/pharmaceutics18020262, *Open Access*.
   Senior author; conceived and supervised the work. Hepatic metabolism and renal excretion coupled to blood pressure regulation; effect of hepatic and renal impairment and genotypes.
2. Nemitz N, Elias M, **König M** (2026). A physiologically based pharmacokinetic and pharmacodynamic (PBPK/PD) model of dapagliflozin in type 2 diabetes mellitus: the effect of dosing, hepatorenal impairment, and food. *Pharmaceutics* 18(3):287. doi:10.3390/pharmaceutics18030287, *Open Access*.
   Senior author; conceived and supervised the work. Shows that renal effects on drug response become visible only in the coupled liver–kidney model.
3. Elias M, **König M** (2025). A digital twin of glimepiride for personalized and stratified diabetes treatment. *Front Pharmacol* 16:1686415. doi:10.3389/fphar.2025.1686415, *Open Access*.
   Senior author; conceived and supervised the work. Patient stratification by hepatic and renal function and CYP2C9 genotype.

4. Höpfl S, Albadry M, Dahmen U, Herrmann KH, Kindler EM, **König M**, Reichenbach JR, Tautenhahn HM, Wei W, Zhao WT, Radde NE (2024). Bayesian modelling of time series data (BayModTS) – a FAIR workflow to process sparse and highly variable data. *Bioinformatics* 40(5):btae312. doi:10.1093/bioinformatics/btae312, *Open Access*.
   Co-author (DFG FOR 5151 QuaLiPerF). Transfers uncertainties of sparse, highly variable time series data into model predictions.
5. Lambers L, Waschinsky N, Schleicher J, **König M**, Tautenhahn HM, Albadry M, Dahmen U, Ricken T (2024). Quantifying fat zonation in liver lobules: an integrated multiscale in silico model combining disturbed microperfusion and fat metabolism via a continuum biomechanical bi-scale, tri-phasic approach. *Biomech Model Mechanobiol*. doi:10.1007/s10237-023-01797-0, *Open Access*.
   Co-author (DFG FOR 5151 QuaLiPerF). Couples microperfusion, oxygen gradient and fat metabolism across scales in a continuum-biomechanical model.
6. Köller A, Grzegorzewski J, Tautenhahn HM, **König M** (2021). Prediction of survival after hepatectomy using a physiologically based pharmacokinetic model of indocyanine green liver function tests. *Front Physiol* 12:730418. doi:10.3389/fphys.2021.730418, *Open Access*.
   Senior author; conceived and supervised the work. Patient-specific prediction of post-operative liver function.
7. Grzegorzewski J, Brandhorst J, Green K, Eleftheriadou D, Duport Y, Barthorscht F, Köller A, Ke DYJ, De Angelis S, **König M** (2021). PK-DB: pharmacokinetics database for individualized and stratified computational modeling. *Nucleic Acids Res* 49(D1):D1358–D1364. doi:10.1093/nar/gkaa990, *Open Access*.
   Senior author; designed and led the database. Open data basis for calibrating and validating digital twins.
8. Sauro HM, …, **König M**, …, Moraru I (2026). From FAIR to CURE: guidelines for computational models of biological systems. *npj Syst Biol Appl*. doi:10.1038/s41540-026-00651-0, *Open Access*.
   Co-author. Community guidelines for credible, understandable, reproducible and extensible models.
9. Keating SM, Waltemath D, **König M**, Zhang F, …, Hucka M, SBML Community (2020). SBML Level 3: an extensible format for the exchange and reuse of biological models. *Mol Syst Biol* 16(8):e9110. doi:10.15252/msb.20199110, *Open Access*.
   Third author; SBML editor and co-author of the specification.
10. **König M**, Bulik S, Holzhütter HG (2012). Quantifying the contribution of the liver to the homeostasis of plasma glucose: a detailed kinetic model of hepatic glucose metabolism integrated with the hormonal control by insulin, glucagon and epinephrine. *PLoS Comput Biol* 8(6):e1002577. doi:10.1371/journal.pcbi.1002577, *Open Access*.
    First author; developed the model. Liver metabolism coupled to whole-body hormonal control.

Category B

1. *Preprint:* Schwaiger A, **König M** (2026). A physiologically based pharmacokinetic/pharmacodynamic model of the diuretic hydrochlorothiazide. *Preprints*. doi:10.20944/preprints202607.0306.v1. *Senior author. Heart, liver and kidney in one model; Open Science Award 2026 of the University of Lübeck.*
2. *Preprint:* Myshkina M, Babaeva M, Nguyen LC, Bedón S, Metternich J, Elias M, **König M** (2026). An open and reproducible digital twin for personalised anticoagulant therapy: modelling apixaban pharmacokinetics and pharmacodynamics. *Preprints*. doi:10.20944/preprints202605.0249.v1. *Senior author.*
3. *Preprint:* Myshkina M, Casabianca E, Schnurpel A, Ricken T, Tautenhahn HM, **König M** (2025). Assessing the impact of AI and digital twins on clinical decision-making in hepatology and hepatobiliary surgery. *Preprints*. doi:10.20944/preprints202509.1164.v1. *Senior author; survey of clinical users.*

4. *Preprint:* Gerhäusser S, Lambers L, Mandl L, Franquinet J, Ricken T, **König M** (2024). Simulation of zonation-function relationships in the liver using coupled multiscale models: application to drug-induced liver injury. *bioRxiv*. doi:10.1101/2024.03.26.586870. *Senior author; coupling of SBML cell models to finite element tissue simulations.*
5. *Reproducible models:* open digital twin models of dapagliflozin and losartan, independently reproduced and published in *Physiome* (2026). doi:10.36903/physiome.31368556, doi:10.36903/physiome.31359823. *Senior author.*
6. *Database:* PK-DB – open pharmacokinetics database, pk-db.com extended to an AI-ready version with an MCP server for AI agents (Anthropic AI for Science Award 2026). *Lead developer.*
7. *Software:* sbmlutils – Python utilities for SBML. doi:10.5281/zenodo.597149. *Lead developer.*
8. *Software:* sbmlsim and SBML4Humans – simulation and human-readable reports of SBML models, sbml4humans.de. *Lead developer.*
9. *Standard:* Smith LP, Bergmann FT, Garny A, Helikar T, Karr J, Nickerson D, Sauro H, Waltemath D, **König M** (2024). The simulation experiment description markup language (SED-ML): language specification for level 1 version 5. *J Integr Bioinform*. doi:10.1515/jib-2024-0008, *Open Access*. *Elected editor; senior author.*
10. *Community roadmap:* EDITH Consortium (2025). Roadmap for the European Virtual Human Twin. doi:10.5281/zenodo.14769224. *Co-author.*

Academic Distinctions

2026	Open Science Award of the University of Lübeck for the hydrochlorothiazide digital twin.
2026	Anthropic AI for Science Award for the project *AI-Ready PK-DB*.
2025	Habilitation completion scholarship, Humboldt-Universität zu Berlin.
2023	Michael Stifel Prize for interdisciplinary, data-driven research, Friedrich Schiller University Jena.
2015	Google Summer of Code scholarship (SBML support for Cytoscape).
2005	Scholarship of the German Academic Scholarship Foundation (Studienstiftung des deutschen Volkes).

Other Information

Involvement in the European digital twin community, directly relevant to TWIN-ORG:

2024–2025	**Co-author of the** *Roadmap for the European Virtual Human Twin* (EDITH Coordination and Support Action, 2025, doi:10.5281/zenodo.14769224). Contributed to the roadmap from the perspective of model standards, reproducibility and FAIR digital twin models.
since 2024	**Member of the VHT user group** of the European Virtual Human Twin (VHT) initiative; participant of the EDITH ecosystem meetings since the first meeting in Paris (01/2024).
since 2025	**Member of the RDA working group** *Building Immune Digital Twins* (Research Data Alliance). Develops standards and methods for immune digital twins; the group contributes an open PBPK digital twin of the immunosuppressant rapamycin (sirolimus).

Third-party funding as principal investigator (selection, since 2016): DFG SPP 2311 SimLivA (second phase 2025–2027), BMFTR ATLAS – AI and Simulation for Tumor Liver Assessment (2023–2026), DFG FOR 5151 QuaLiPerF (2021–2025), EU EOSC-Life (2020–2021), BMBF LiSyM junior group (2016–2022). All models, data and software of the group are published openly and in community standards (SBML, SED-ML) to make them reusable across groups.